

Love AI More. Trust AI Less.

DSCT lecture walkthrough

Abstract

A short report can be a reasoning instrument: state a claim, inspect evidence, and decide how much confidence is earned.

1 The question

Claim: Increasing simultaneous requests should increase aggregate AI throughput until parallelism is saturated. This matters because “bigger GPU is faster” can confuse capacity with scheduling.

2 Evidence: a generated table

The cells report aggregate tokens per second / overlap factor. The overlap factor is aggregate rate divided by the median per-request generation rate.

Hardware	parallel limit	$N = 1$	$N = 4$	$N = 8$	$N = 32$
Tesla T4	1	69.2 / 0.49	74.1 / 0.54	73.4 / 0.53	73.4 / 0.54
Tesla T4	8	72.6 / 0.53	102.8 / 2.18	117.3 / 4.77	121.9 / 4.84
GTX 1060 3 GB	1	59.6 / 0.61	84.5 / 0.86	90.2 / 0.92	94.6 / 0.98
GTX 1060 3 GB	4	73.6 / 0.99	112.1 / 3.76	120.7 / 3.98	119.9 / 3.97

Each cell: aggregate tokens/s / overlap factor.